



SAINT IGNATIUS' COLLEGE

Trial Higher School Certificate

2007

MATHEMATICS EXTENSION 2

12:30pm – 3:35pm

Tuesday 28th August 2007

Directions to Students

• Reading Time : 5 minutes	• Total Marks 120
• Working Time : 3 hours	
• Write using blue or black pen. (sketches in pencil).	• Attempt Question 1 – 8
• Board approved calculators may be used	• All questions are of equal value
• A table of standard integrals is provided at the back of this paper.	
• All necessary working should be shown in every question.	
• Answer each question in the booklets provided and clearly label your name and teacher's name.	

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Question 1 (15 marks) Use a SEPARATE writing booklet

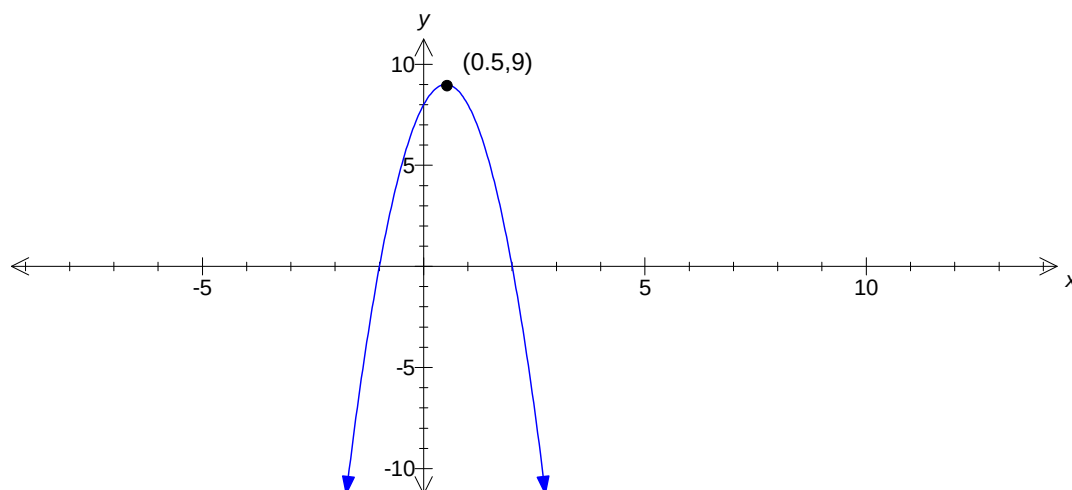
Marks

- (a) (i) Show that $\sin(A+B) + \sin(A-B) = 2 \sin A \cos B$ 1
- (ii) Hence find the indefinite integral $\int \sin 5x \cos 3x \, dx$ 2
- (b) Evaluate $\int_0^5 \frac{t \, dt}{\sqrt{t+4}}$ 2
- (c) Evaluate $\int_{-1}^1 3^x \, dx$ correct to three significant figures. 2
- (d) Evaluate $\int_0^{\frac{1}{3}} \frac{dx}{\sqrt{1-9x^2}}$ 2
- (e) Find $\int \frac{1}{1+\sin x} \, dx$, using the substitution $t = \tan \frac{x}{2}$ 3
- (f) Use integration by parts to find $\int \frac{\cos^{-1} x}{\sqrt{1+x}} \, dx$ 3

Question 2 (15 marks) Use a SEPARATE writing booklet**Marks**

- (a) Prove that $f(x) = \frac{x^3}{\sin x}$ is an even function. 2

- (b) A sketch of $f(x) = -4(x+1)(x-2)$ is shown below.



With the aid of the above diagram, and without the use of calculus, draw a separate half page sketch for each of the following.

- (i) $y = |f(x)|$ 1
- (ii) $y = f(2x)$ 1
- (iii) $y = f(-x)$ 1
- (iv) $y = \frac{1}{f(x)}$ 2
- (v) $y = \sqrt{f(x)}$ 2
- (vi) $y = \log_e f(x)$ 2
- (c) Using calculus, show that $e^{-x} + x - 1 \geq 0$ for real x . 4

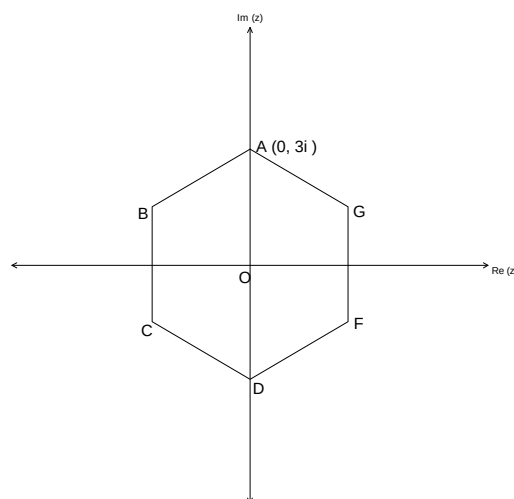
Question 3 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) Express the following in the form $(x+iy)$, where x and y are real: 2

$$\frac{i^2-1}{i} + \frac{1}{1+i}$$

- (b)



The Argand diagram above, shows a regular hexagon with vertex A at the point $(0, 3i)$. O is the centre of the hexagon.

- (i) Copy the diagram into your writing booklet.
- (ii) On your diagram show the region within the hexagon in which both the inequalities $|z| \leq 2$ and $-\frac{\pi}{6} \leq \arg z \leq \frac{\pi}{6}$ are satisfied. 2
- (iii) Find in the form $|z - z_1| = R$, the equation of the circle through the points O , B and F . 1
- (iv) Find the complex numbers, in modulus argument form, represented by the points B and C . 2
- (v) The hexagon is rotated anticlockwise about the origin through an angle of $\frac{\pi}{4}$. Express in the form $r(\cos \theta + i \sin \theta)$, where θ is the principal argument, the complex numbers represented by the new positions of B and C . 3

Question 3 continues on page 5

Question 3 (continued)

- (c) If $1 - 2i$ is a root of the equation $z^2 - (3 + i)z + k = 0$,
- (i) explain why the conjugate $1 + 2i$ cannot be a root to the equation 1
 - (ii) show that the other root is $2 + 3i$ 1
 - (iii) find the value of k 1
 - (iv) hence, or otherwise, find the two square roots of $-24 + 10i$. 2

Question 4 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) A polynomial $p(x) = x^n + ax^2 - 2$ has a factor of $(x-1)$ and leaves a remainder of -6 on division by $(x+2)$.
Find:
- (i) the value of a 1
 - (ii) the value of n 1
 - (iii) the zeros of $p(x)$. 2
- (b) Find the values of a and b so that $p(x) = 2x^3 - (2a+1)x^2 + (2+b)x - 1$ has a double root at $x = 1$. 4
- (c) If l, m, n are the roots of the equation $x^3 - 2x + 5 = 0$,
- (i) find the cubic equation whose roots are $2l, 2m, 2n$. 2
 - (ii) find the value of $l^3 + m^3 + n^3$. 2
- (d) Find all the values of k for which the polynomial equation $3x^4 - 4x^3 + k = 0$ has no real roots. 3

Question 5 (15 marks) Use a SEPARATE writing booklet**Marks**

- (a) An ellipse has the equation $\frac{x^2}{25} + \frac{y^2}{16} = 1$. O is the centre of the ellipse and S and S' are the foci.
- (i) Find
- (α) the eccentricity 1
 - (β) the co-ordinates of the foci 1
 - (γ) the equations of the directrices. 1
- (ii) Make a third of a page sketch of the ellipse showing the features found in part (i) 2
- (iii) If $P(x_0, y_0)$ is a point on the ellipse show that $(PS + PS')$ is constant. You may mark point P in quadrant one of the above mentioned diagram. 2
- (iv) Show that the equation of the tangent at $P(x_0, y_0)$ is 2
- $$\frac{xx_0}{25} + \frac{yy_0}{16} = 1.$$
- (v) The tangent at P meets the nearer focus at R . If S is the nearer focus to P ,
- (α) write down the co-ordinates of R . 1
 - (β) find expressions for the gradients of PR and SR in terms of x_0 and y_0 . 2
 - (γ) show that the angle PSR is a right angle. 1
- (b) A hyperbola has its centre at the origin and asymptotes $y = \pm \frac{2}{3}x$. Find its equation. 2

Question 6 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) The region bounded by the curve $y = x(2 - x)$ and the x -axis is rotated about the y -axis. Find the volume of the solid of revolution by taking slices perpendicular to the y -axis. 4
- (b) The region bounded by the curve $y = \ln x$, the line $y = 1$ and the co-ordinate axes is rotated about the x -axis.
- (i) By dividing the resulting solid into cylindrical shells, show that each shell has an approximate volume : $\delta v = 2\pi y e^y \delta y$ where δy is the thickness of the shell. 2
- (ii) Hence calculate the volume of the solid. 2
- (c) The base of a particular solid is the circle $x^2 + y^2 = 8$. Find the volume of the solid if every cross section to the x -axis is an isosceles – right angled triangle with the hypotenuse in the base of the solid. 4
- (d) Show that the straight line $lx + my + n = 0$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ if $a^2 l^2 - b^2 m^2 = n^2$. 3

Question 7 (15 marks) Use a SEPARATE writing booklet**Marks**

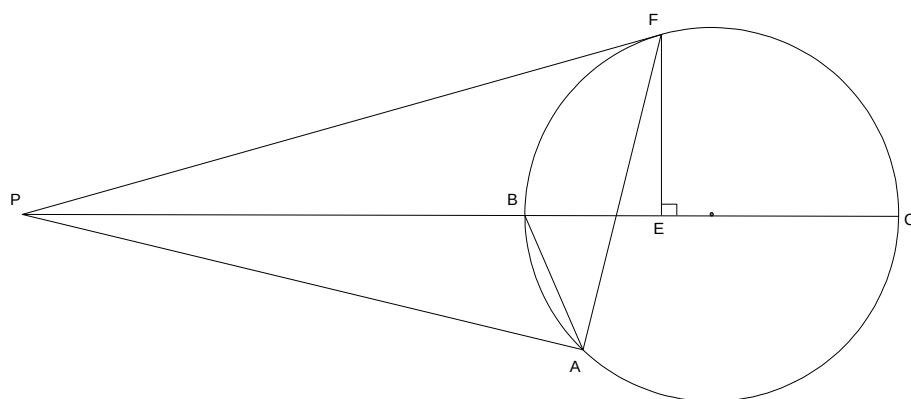
- (a) Mr Kirkpatrick's mathematics class brought him a ride in a Gondola at Queenstown in New Zealand, during a recent trip. When Mr Kirkpatrick's hands were H metres above the Earth's surface, he dropped overboard his packet of beer nuts of mass m kg. The packet of beer nuts encounters air resistance proportional to its velocity v (which is in metres per second), that is the resistive force is equal to mkv .
Taking Mr Kirkpatrick's hands as the origin and downwards displacement as positive:
- (i) Write down an equation of motion representing the passage of the packet of beer nuts. 2
- (ii) Find the terminal velocity, w , of the packet of beer nuts. 1
- (iii) Show that the equation of motion in part (i) can be written as $\ddot{x} = k(w - v)$. 1
- (iv) Show that the displacement, x metres, of the packet of beer nuts from Mr Kirkpatrick's hands is given by: $x = -\frac{v}{k} - \frac{w}{k} \ln\left(\frac{w-v}{w}\right)$. 4
- (v) If the packet reaches the Earth's surface with a velocity of u metres per second, show that $\ln\left(1 - \frac{u}{w}\right) + \frac{u}{w} + \frac{kH}{w} = 0$. 1
- (vi) Consider the moment when the packet of beer nuts has reached 75% of its terminal velocity.
Find:
- (α) the time, t seconds, for this moment to be reached. 3
- (β) the distance fallen at this moment. 3

Question 8 (15 marks) Use a SEPARATE writing booklet

Marks

(a) Draw a neat half page sketch of the graph for $y^2 = x^2(4 - x^2)$. 3

(b)



In the above diagram AB and BC are chords of a circle, and F is on the arc ABC such that arc AF is equal to arc FC . E is the foot of the perpendicular from F to the chord BC . CB is extended to P so that $PE = EC$. (Note that B is inside the triangle APF)

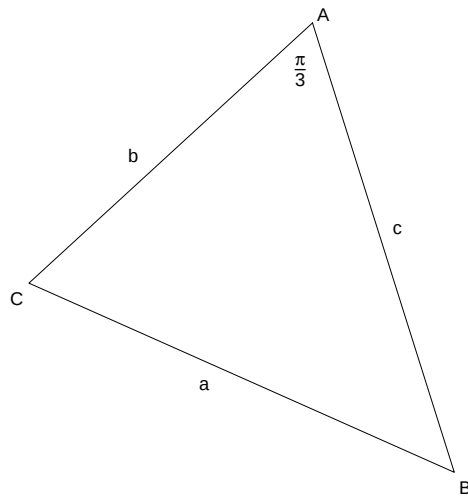
(i) Show that the triangle APF is isosceles. 3

(ii) Show that $AB + BE = EC$. 4

Question 8 continues on page 11

Question 8 (continued)

(c)



A triangle ABC has sides of varying length a , b and c with a fixed interior angle of $BAC = \frac{\pi}{3}$ as shown in the above diagram.

Use the cosine rule to show that:

- | | | |
|------|---|---|
| (i) | $a^2 \geq bc$, and hence, | 3 |
| (ii) | the area of triangle $ABC \leq \frac{a^2\sqrt{3}}{4}$ | 2 |

End of examination

Solutions



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MATHEMATICS EXTENSION 2

- This is a study resource for HSC preparation
- This document is designed to help students understand the questions
- The solutions should be treated as aids only
- There may be better solutions to some questions.

Question 1

$$(a)(i) LHS = \sin(A+B) + \sin(A-B)$$

$$= \sin A \cos B + \cos A \sin B + \sin A \cos B - \cos A \sin B$$

$$= 2 \sin A \cos B = RHS$$

$$(ii) \int \sin 5x \cos 3x dx = \frac{1}{2} \int (\sin 8x + \sin 2x) dx$$

$$= \frac{1}{2} \left(-\frac{1}{8} \cos 8x - \frac{1}{2} \cos 2x \right) + C$$

$$= -\frac{1}{16} \cos 8x - \frac{1}{4} \cos 2x + C$$

$$(b) I = \int_0^5 \frac{t dt}{\sqrt{t+4}}$$

$$\text{Let } u = t+4$$

$$du = dt$$

$$\text{When } t=5, u=9$$

$$t=0, u=4$$

$$I = \int_4^9 \frac{u-4}{u^{1/2}} du$$

$$I = \int_4^9 \left(u^{\frac{1}{2}} - 4u^{-\frac{1}{2}} \right) du$$

$$I = \left[\frac{2}{3} u^{\frac{3}{2}} - 2 \times 4 u^{\frac{1}{2}} \right]_4^9$$

$$I = \left[\frac{2}{3} u^{\frac{3}{2}} - 8u^{\frac{1}{2}} \right]_4^9$$

$$I = \left[\left(\frac{2}{3} \times 27 - 24 \right) - \left(\frac{2}{3} \times 8 - 8 \times 2 \right) \right]$$

$$I = 18 - 24 - \frac{16}{3} + 16$$

$$I = 34 - 29\frac{1}{3}$$

$$I = 4\frac{2}{3}$$

$$\begin{aligned} (c) \quad \int_{-1}^1 3^x dx &= \frac{1}{\ln 3} [3^x]_{-1}^1 \\ &= \frac{1}{\ln 3} \left(3 - \frac{1}{3} \right) \\ &= \frac{8}{3 \ln 3} \end{aligned}$$

$$= 2.43 \text{ c.s.f.} \quad \text{Calculator Face } 2.427304604$$

$$(d) \quad \int_0^{\frac{1}{3}} \frac{dx}{\sqrt{1-9x^2}}$$

Note $1-9x^2 = 9\left(\frac{1}{9}-x^2\right)$

$$\sqrt{1-9x^2} = 3\sqrt{\frac{1}{9}-x^2}$$

$$= \frac{1}{3} \int_0^{\frac{1}{3}} \frac{1}{\sqrt{\left(\frac{1}{3}\right)^2 - x^2}} dx$$

$$= \frac{1}{3} \left[\sin^{-1} \frac{x}{\frac{1}{3}} \right]_0^{\frac{1}{3}}$$

$$= \frac{1}{3} \left[\sin^{-1} 3x \right]_0^{\frac{1}{3}}$$

$$= \frac{1}{3} [\sin^{-1} 1 - \sin^{-1} 0]$$

$$= \frac{1}{3} \times \frac{\pi}{2}$$

$$= \frac{\pi}{6}$$

$$(e) \int \frac{1}{1+\sin x} dx$$

$$\text{when } t = \tan \frac{x}{2}$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$2 dt = (1 + \tan^2 \frac{x}{2}) dx$$

$$\frac{2 dt}{1+t^2} = dx$$

$$I = \int \frac{1}{1 + \frac{2t}{1+t^2}} \cdot \frac{2 dt}{1+t^2}$$

$$I = 2 \int \frac{1+t^2}{(1+t^2+2t)(1+t^2)} dt$$

$$I = 2 \int \frac{1}{(1+t)^2} dt$$

$$I = 2 \int (1+t)^{-2} dt$$

$$I = 2 \frac{(1+t)^{-1}}{-1} + C$$

$$I = -\frac{2}{1+t} + C$$

$$I = -\frac{2}{1+\tan \frac{x}{2}} + C$$

$$(f) \int \frac{\cos^{-1} x}{\sqrt{1+x}} dx$$

$$\text{Let } u = \cos^{-1} x \Rightarrow u' = -\frac{1}{\sqrt{1-x^2}}$$

$$v' = (1+x)^{-\frac{1}{2}} \Rightarrow v = \frac{(x+1)^{\frac{1}{2}}}{\frac{1}{2} \times 1} = 2\sqrt{x+1}$$

$$I = 2\sqrt{x+1} \cos^{-1} x - \int -\frac{1}{\sqrt{1-x^2}} \times 2\sqrt{x+1} dx$$

$$I = 2\sqrt{x+1} \cos^{-1} x + 2 \int \frac{\sqrt{x+1}}{\sqrt{1-x^2}} dx$$

$$I = 2\sqrt{x+1} \cos^{-1} x + 2 \int \frac{1}{\sqrt{1-x}} dx$$

$$I = 2\sqrt{x+1} \cos^{-1} x + 2 \int (1-x)^{-\frac{1}{2}} dx$$

$$I = 2\sqrt{x+1} \cos^{-1} x + 2 \times \frac{(1-x)^{\frac{1}{2}}}{\frac{1}{2} \times -1} + C$$

$$I = 2\sqrt{x+1} \cos^{-1} x - 4\sqrt{1-x} + C$$

Question 2

(a) $f(x) = \frac{x^3}{\sin x}$

$$f(-x) = \frac{(-x)^3}{\sin(-x)}$$

$$= \frac{-x^3}{-\sin x}$$

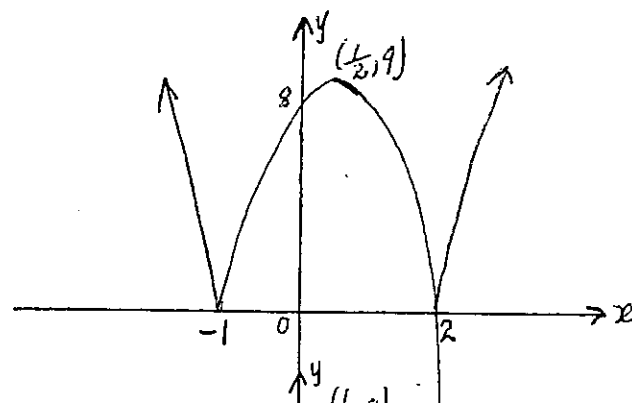
$$= \frac{x^3}{\sin x}$$

$$= f(x)$$

$\therefore f(x)$ is an even function

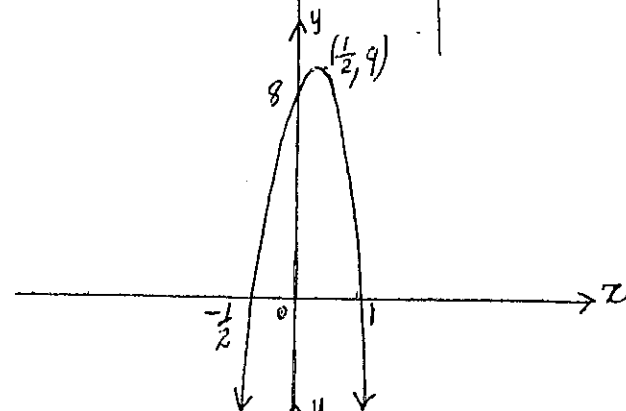
(b) $f(x) = -4(x+1)(x-2)$

(i)



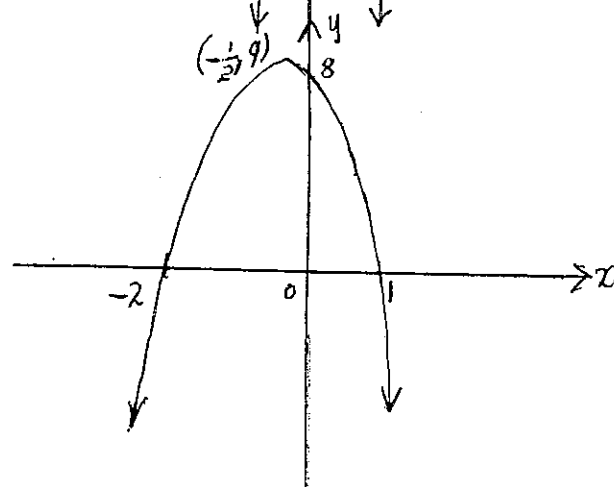
$y = |f(x)|$

(ii)



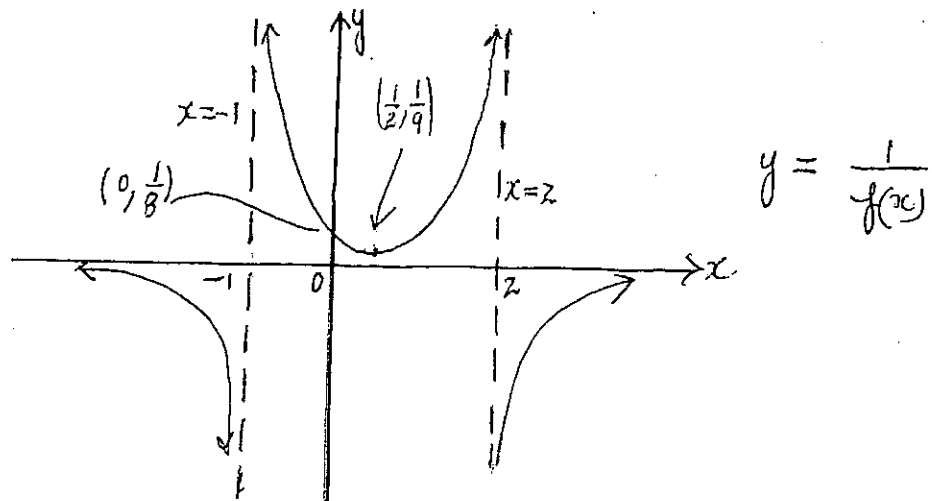
$y = f(2x)$

(iii)

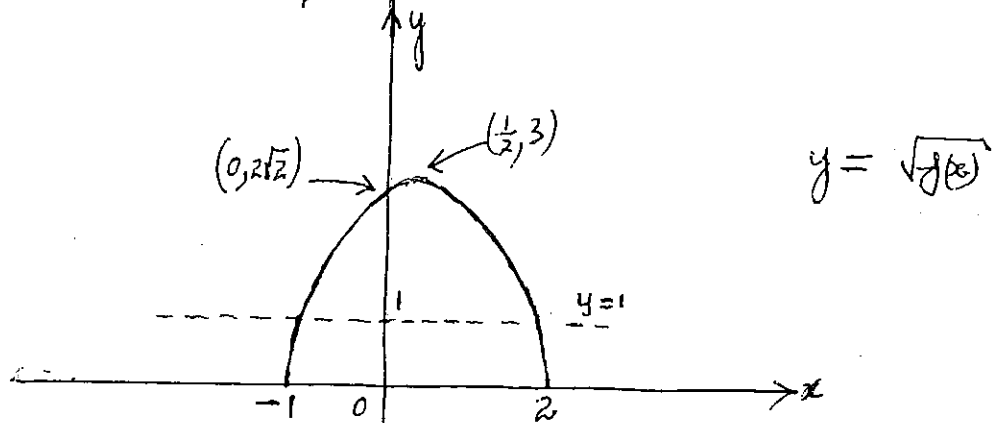


$y = f(-x)$

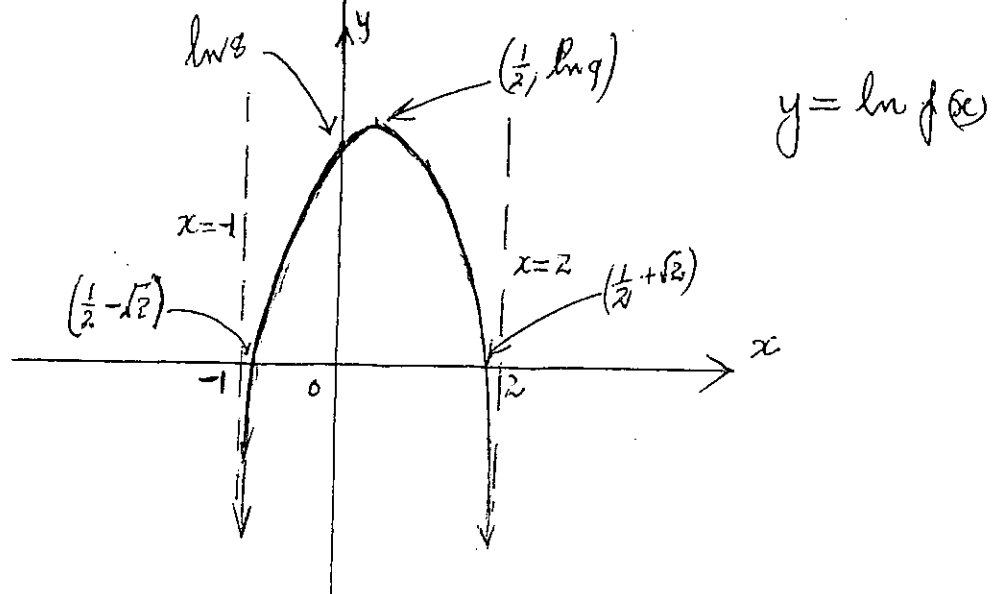
(iv)



(v)



(vi)



$$(c) \quad e^{-x} + x - 1 \geq 0$$

$$\text{Let } y = e^{-x} + x - 1$$

$$\text{Now } \frac{dy}{dx} = -e^{-x} + 1$$

$$\frac{d^2y}{dx^2} = e^{-x} = \frac{1}{e^x} > 0 \text{ for all } x \text{ in the domain of the reals.}$$

Note: The meaning of a negative index
 $a^{-m} = \frac{1}{a^m}$

Now since $\frac{d^2y}{dx^2} > 0$ we can say that the curve.

$y = e^{-x} + x - 1$ is always concave up.

$$\text{When } \frac{dy}{dx} = 0$$

$$-e^{-x} + 1 = 0$$

$$e^x = 1$$

$$e^x = e^0$$

$$\therefore x = 0$$

$$\text{and when } x = 0, y = e^0 + 0 - 1$$

$$y = 0$$

$\therefore$ minimum turning point at $(0, 0)$

Hence $y \geq 0$ for all x hence

$$e^{-x} + x - 1 \geq 0 \text{ for real } x.$$

(C) $(1-2i)$ a root of $z^2 - (3+i)z + k = 0$

(I) Theorem: If polynomial equation co-efficients are real then complex roots exist in conjugate pairs. Hence $(1+2i)$ is not a root.

(II) Sum of roots: $\alpha + \beta = -\frac{[-(3+i)]}{1}$

$$\alpha + \beta = 3+i$$

If $\alpha = 1-2i$

Then $1-2i + \beta = 3+i$

$$\therefore \beta = 2+3i$$

(III) Product of roots: $\alpha\beta = \frac{k}{1}$

$$\alpha\beta = k$$

substitution: $(1-2i)(2+3i) = k$

$$k = 2 + 3i - 4i - 6i^2$$

$$k = 2 + 6 - i$$

$$k = 8 - i$$

(IV) Now considering $z^2 - (3+i)z + (8-i) = 0$

$$z = \frac{(3+i) \pm \sqrt{(3+i)^2 - 4(1)(8-i)}}{2(1)}$$

$$z = \frac{(3+i) \pm \sqrt{9+6i+i^2-32+4i}}{2}$$

$$z = \frac{(3+i) \pm \sqrt{-24+10i}}{2}$$

Note here that
There is an
alternative
method for
finding square
roots

Let one of the square roots be $(a+bi)$

$$\therefore 1-2i = \frac{(3+i) + a+bi}{2}$$

$$2-4i = 3+i+a+bi$$

$$-1-5i = a+bi$$

Equating real and imaginary parts.

$$a = -1, b = -5$$

Similarly it can also be shown that $a=1, b=5$

$\therefore$ The two square roots are $\pm(1+5i)$

Question 4

(a) $p(x) = x^n + ax^2 - 2$

(i) When $x=1$, $p(x) = 0$

$\therefore 0 = (1)^n + a(1)^2 - 2$

$\therefore a = 1$

(ii) When $p(x) = -6$, $x = -2$

$\therefore -6 = (-2)^n + 1(-2)^2 - 2$

$-6 = (-2)^n + 4 - 2$

$-6 = (-2)^n + 2$

$(-2)^n = -8$

$(-2)^n = (-2)^3$

$n = 3$

(iii) $\therefore p(x) = x^3 + x^2 - 2$

Now $p(x) \div (x-1)$

$$\begin{array}{r} x-1 \overline{) \begin{array}{r} x^3 + x^2 + 0x - 2 \\ \underline{x^3 - x^2} \\ 2x^2 + 0x \\ \underline{2x^2 - 2x} \\ 2x - 2 \\ \underline{2x - 2} \\ 0 + 0 \end{array}} \end{array}$$

$\therefore p(x) = (x-1)(x^2 + 2x + 2)$

When $x^2 + 2x + 2 = 0$

$x = \frac{-2 \pm \sqrt{4-8}}{2}$

$x = \frac{-2 \pm \sqrt{4-8}}{2}$

$x = \frac{-2 \pm 2i}{2} = -1 \pm i$

Hence solutions for $p(x) = 0$ are $x = 1$ or $x = -1 \pm i$

$$(b) \quad p(x) = 2x^3 - (2a+1)x^2 + (2+b)x - 1$$

$$p'(x) = 6x^2 - 2(2a+1)x + (2+b)$$

Now There is a double root at $x=1$

$$\therefore p(1) = p'(1) = 0$$

$$\text{So } 2 - (2a+1) \times 1 + (2+b) \times 1 - 1 = 0$$

$$2 - 2a - 1 + 2 + b - 1 = 0$$

$$-2a + b + 2 = 0$$

$$2a - b = 2 \quad \text{--- (1)}$$

$$\text{And } 6 - 2(2a+1) \times 1 + 2 + b = 0$$

$$6 - 4a - 2 + 2 + b = 0$$

$$-4a + b + 6 = 0$$

$$4a - b = 6 \quad \text{--- (2)}$$

$$\text{Solving: } (1) - (2) \quad -2a = -4$$

$$a = 2$$

$$\text{Sub in (1)} \quad 4 - b = 2$$

$$b = 2$$

$$\text{Answer } a=2, b=2$$

$$(b) \quad l, m, n \text{ roots of } x^3 - 2x + 5 = 0$$

$$(1) \quad \text{Let } y = 2x \Rightarrow x = \frac{y}{2}$$

Substitute:

$$\left(\frac{y}{2}\right)^3 - 2\left(\frac{y}{2}\right) + 5 = 0$$

$$\frac{y^3}{8} - y + 5 = 0$$

$$y^3 - 8y + 40 = 0$$

reverting to the variable x :

$$\text{The required equation is } x^3 - 8x + 40 = 0$$

C (ii) If l, m, n are the roots of the equation $x^3 = 2x - 5$ Then:

$$l^3 = 2l - 5 \quad \text{--- ①}$$

$$m^3 = 2m - 5 \quad \text{--- ②}$$

$$n^3 = 2n - 5 \quad \text{--- ③}$$

Add these equations:

$$l^3 + m^3 + n^3 = 2(l + m + n) - (5 \times 3)$$

Now the sum of the roots of the given equation is zero, i.e. $(l + m + n) = 0$

$$\therefore l^3 + m^3 + n^3 = -15$$

(d) considering $3x^4 - 4x^3 + k = 0$.

$$\text{Let } p(x) = 3x^4 - 4x^3 + k.$$

$$p'(x) = 12x^3 - 12x^2 = 12x^2(x - 1)$$

$$p''(x) = 36x^2 - 24x = 12x(3x - 2)$$

For stationary points $p'(x) = 0 \therefore x = 0$ or 1

When $x = 0$ $p(x) = k$

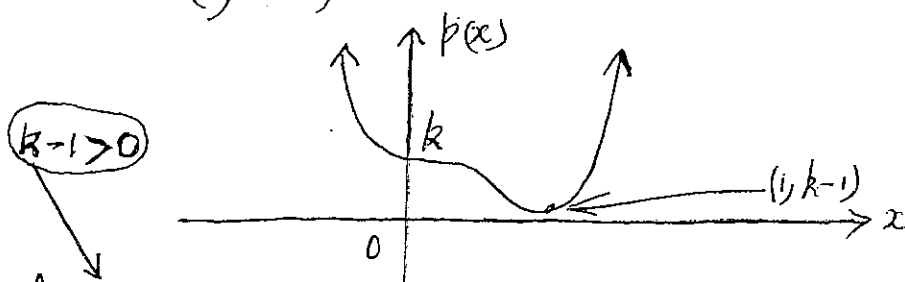
and $p''(x) = 0$

It appears that $(0, k)$ is a horizontal point of inflection

When $x = 1$ $p(x) = k - 1$

and $p''(x) = 12 > 0$

$\therefore (1, k - 1)$ is a minimum turning point.



If $k > 1$ The point $(1, k - 1)$ is above the x-axis and hence $(3x^4 - 4x^3 + k) = 0$ has no real roots.

Question 5

(a) $\frac{x^2}{25} - \frac{y^2}{16} = 1$

$a = 5, b = 4$

(i) $b^2 = a^2(e^2 - 1)$

$16 = 25(e^2 - 1)$

$\frac{16}{25} = e^2 - 1$

$e^2 = \frac{25}{25} + \frac{16}{25}$

$e^2 = \frac{41}{25}$

$e = \frac{\sqrt{41}}{5}, e > 0$

(ii) $S \equiv (ae, 0); S' \equiv (-ae, 0)$

$ae = \frac{5 \times \sqrt{41}}{5} = \sqrt{41}$

$\therefore S = (\sqrt{41}, 0) \text{ and } S' = (-\sqrt{41}, 0)$

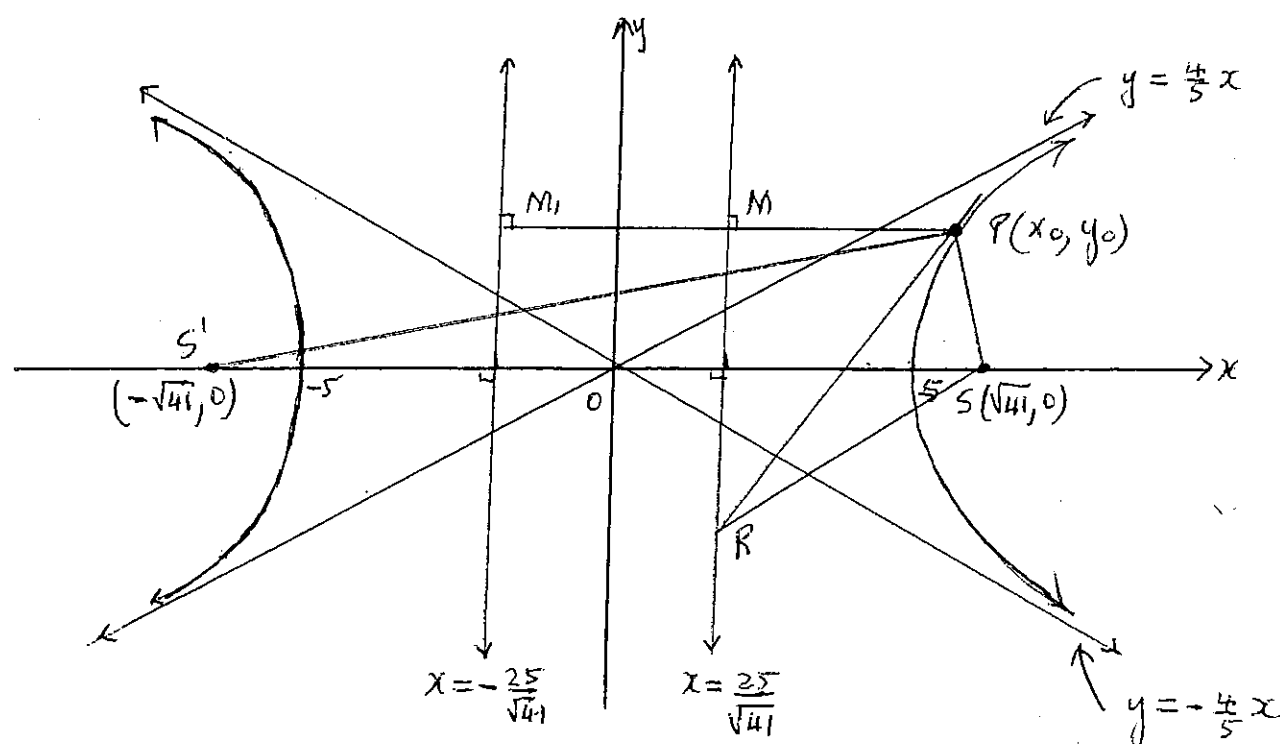
(iii) $y = \pm \frac{b}{a}x$

$y = \pm \frac{4}{5}x$

(iv) $x = \pm \frac{a}{e}$

$x = \pm \frac{25}{\sqrt{41}}$

(b)



(c) $\frac{PS}{PM} = e$ and $\frac{PS'}{PM'} = e$ This is from the locus definition of a Hyperbola.

$$PS - PS' = ePM - ePM'$$

$$= e(PM - PM')$$

$$|PS - PS'| = \left| \frac{\sqrt{41}}{5} \left[x_0 - \frac{25}{\sqrt{41}} - \left(x_0 + \frac{25}{\sqrt{41}} \right) \right] \right|$$

$$= \left| \frac{\sqrt{41}}{5} \left(x_0 - \frac{25}{\sqrt{41}} - x_0 - \frac{25}{\sqrt{41}} \right) \right|$$

$$= \left| \frac{\sqrt{41}}{5} \left(-\frac{50}{\sqrt{41}} \right) \right|$$

$$= 10 \text{ i.e. a constant (This is the length of the major axis).}$$

(d) Equation of tangent:

$$\frac{2x}{25} - \frac{4}{8} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{16x}{25y}$$

Gradient of tangent: $m = \frac{16x_0}{25y_0}$

tangent: $y - y_1 = m(x - x_1)$

$$y - y_0 = \frac{16x_0}{25y_0} (x - x_0)$$

$$25y_0y - 25y_0^2 = 16x_0x - 16x_0^2$$

$$16x_0x - 25y_0y = 16x_0^2 - 25y_0^2$$

$$\frac{x_0x}{25} - \frac{y_0y}{16} = \frac{x_0^2}{25} - \frac{y_0^2}{16} \quad \neq$$

$$\text{i.e. } \frac{x_0x}{25} - \frac{y_0y}{16} = 1 \quad \text{--- --- --- } \textcircled{1}$$

Note: (x_0, y_0) satisfies the equation of the hyperbola

Hence $\frac{x_0^2}{25} - \frac{y_0^2}{16} = 1$

(e) (i) Solving ① with $x_R = \frac{25}{\sqrt{41}}$ --- ② for the co-ordinates of R.

$$\text{Sub ② in ①} \quad \frac{x_0}{25} \times \frac{25}{\sqrt{41}} - \frac{y_0}{16} = 1$$

$$\frac{x_0}{\sqrt{41}} - 1 = \frac{y_0}{16}$$

$$y_R = \frac{16}{y_0} \left(\frac{x_0}{\sqrt{41}} - 1 \right)$$

The co-ordinates of R are given by (x_R, y_R)

(ii) Gradients : $m = \frac{y_2 - y_1}{x_2 - x_1}$

$$m_{PS} = \frac{y_0 - 0}{x_0 - \sqrt{41}} = \frac{y_0}{x_0 - \sqrt{41}}$$

$$m_{SR} = \frac{\frac{16}{y_0} \left(\frac{x_0}{\sqrt{41}} - 1 \right) - 0}{\frac{25}{\sqrt{41}} - \sqrt{41}} = \frac{\frac{16}{y_0 \sqrt{41}} (x_0 - \sqrt{41})}{-\frac{16}{\sqrt{41}}}$$

(iii) For perpendicular lines $m_1 m_2 = -1$

$$m_{PS} \times m_{SR} = \frac{y_0}{x_0 - \sqrt{41}} \times \frac{\sqrt{41}}{-16} \times \frac{16}{y_0 \sqrt{41}} (x_0 - \sqrt{41})$$

$$= -1$$

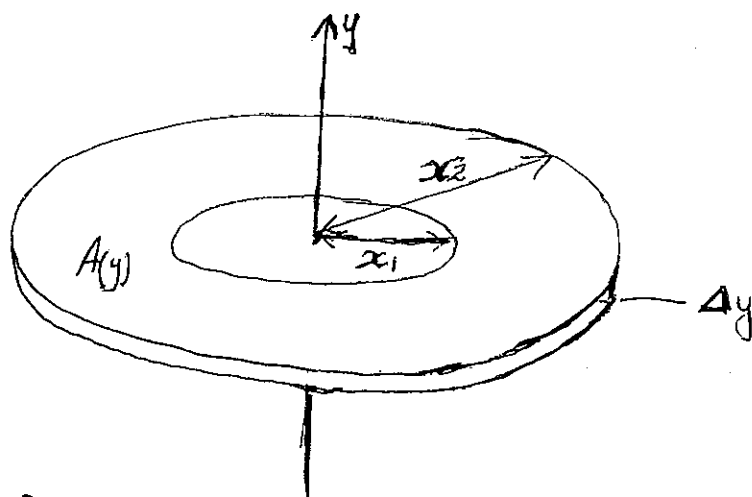
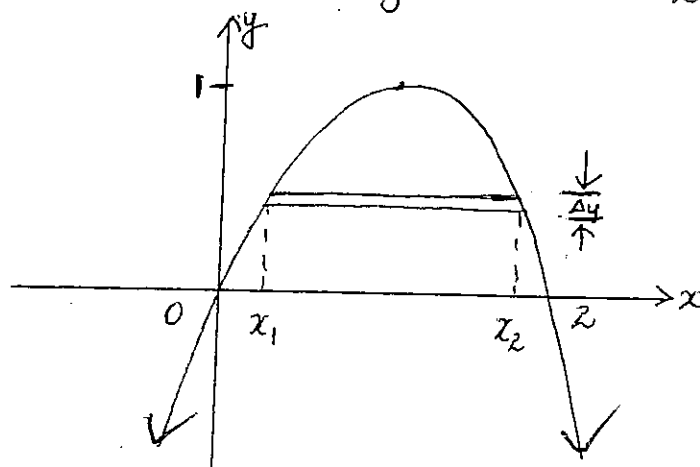
Therefore $PS \perp SR$

Hence $\widehat{PSR} = 90^\circ$ i.e. a right angle.

Question 6

(a)

$$y = 2x - x^2 = x(2-x)$$



$$A(y) = \pi x_2^2 - \pi x_1^2$$

$$\Delta V \doteq \pi(x_2^2 - x_1^2) \Delta y.$$

$$V = \lim_{\Delta y \rightarrow 0} \int_{y=0}^{y=1} \pi x 4 \sqrt{1-y} \Delta y$$

$$V = 4\pi \int_0^1 (1-y)^{\frac{1}{2}} dy$$

$$V = 4\pi \left[\frac{(1-y)^{\frac{3}{2}}}{\frac{3}{2} \times -1} \right]_0^1$$

$$V = 4\pi \times -\frac{2}{3} [0 - 1]$$

$$V = \frac{8\pi}{3} \text{ units}^3$$

Note To find x_1 and x_2

Solve the quadratic equation in x where y is a fixed value

$$x^2 - 2x + y = 0$$

$$x = \frac{2 \pm \sqrt{4-4y}}{2}$$

$$x = 1 \pm \sqrt{1-y}$$

$$x_2 = 1 + \sqrt{1-y}$$

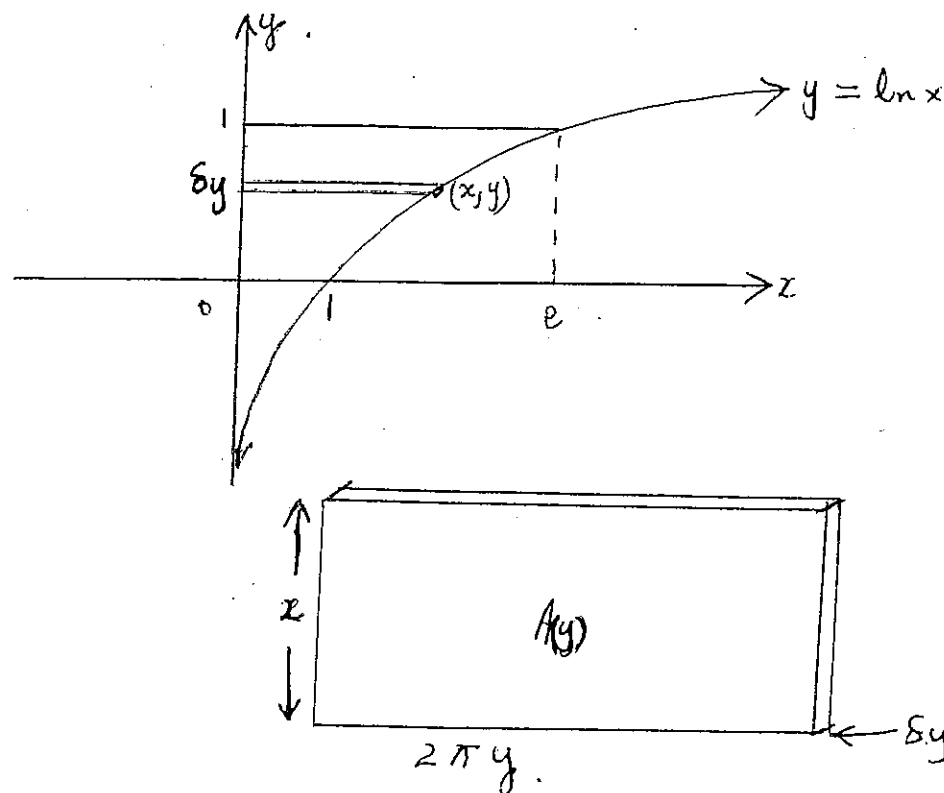
$$x_1 = 1 - \sqrt{1-y}$$

And $x_2^2 - x_1^2 = (x_2 + x_1)(x_2 - x_1)$

$$= 2(2\sqrt{1-y})$$

$$= 4\sqrt{1-y}$$

(b)



$$A(y) = 2\pi y \times x = 2\pi xy$$

$$V \doteq 2\pi xy \Delta y$$

$$\delta V \doteq 2\pi x e^y xy \delta y$$

$$\text{i.e. } \delta V \doteq 2\pi y e^y \delta y$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{y=0}^{y=1} 2\pi y e^y \delta y$$

$$V = 2\pi \int_0^1 y e^y dy$$

Now integration by parts with $u=y \Rightarrow u'=1$
 $v'=e^y \Rightarrow v=e^y$

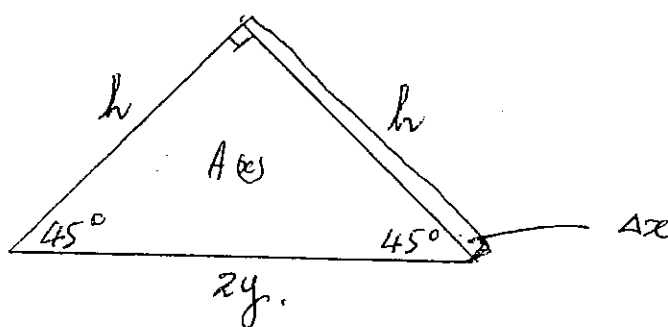
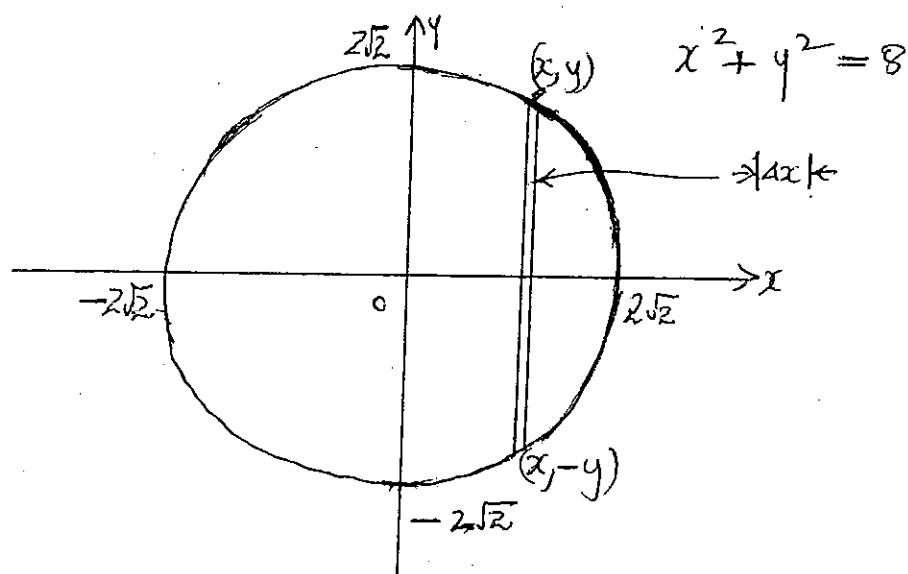
$$V = 2\pi \left\{ \left[y e^y \right]_0^1 - \int_0^1 e^y dy \right\}$$

$$V = 2\pi \left\{ (e - 0) - \left[e^y \right]_0^1 \right\}$$

$$V = 2\pi (e - e + 1)$$

$$V = 2\pi \text{ units}^3$$

(c)



$$\text{Now } \frac{h}{2y} = \cos 45^\circ = \frac{1}{\sqrt{2}} \Rightarrow h = \frac{2y}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = y\sqrt{2}$$

$$A(x) = \frac{b \times h}{2} = \frac{y\sqrt{2} \times y\sqrt{2}}{2} = y^2$$

$$\Delta V \doteq y^2 \Delta x$$

$$\Delta V \doteq (8 - x^2) \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=-2\sqrt{2}}^{x=2\sqrt{2}} (8 - x^2) \Delta x$$

$$V = 2 \int_0^{2\sqrt{2}} (8 - x^2) dx$$

note even function property.

$$V = 2 \left[8x - \frac{x^3}{3} \right]_0^{2\sqrt{2}}$$

$$V = 2 \left[16\sqrt{2} - \frac{16\sqrt{2}}{3} - 0 \right]$$

$$V = \frac{64\sqrt{2}}{3} \text{ units}^3$$

$$(d) \quad lx + my + n = 0 \quad \text{--- --- --- (1)}$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{--- --- --- (2)}$$

$$\text{From (1)} \quad y = -\frac{lx+n}{m}$$

Sub in (2)

$$\frac{x^2}{a^2} + \frac{(lx+n)^2}{b^2m^2} = 1$$

$$b^2m^2x^2 + a^2(l^2x^2 + 2lnx + n^2) = a^2b^2m^2$$

$$b^2m^2x^2 + a^2l^2x^2 + 2a^2lnx + a^2n^2 - a^2b^2m^2 = 0$$

$$(b^2m^2 + a^2l^2)x^2 + 2a^2lnx + a^2(n^2 - b^2m^2) = 0$$

In this quadratic equation $\Delta = 0$ for tangency

$$(2a^2ln)^2 - 4(b^2m^2 + a^2l^2) \times a^2(n^2 - b^2m^2) = 0$$

$$\div 4a^2 : a^2l^2n^2 - (b^2m^2 + a^2l^2)(n^2 - b^2m^2) = 0$$

$$a^2l^2n^2 - (b^2m^2n^2 - b^4m^4 + a^2l^2n^2 - a^2l^2b^2m^2) = 0$$

$$a^2l^2n^2 - b^2m^2n^2 + b^4m^4 - a^2l^2n^2 + a^2l^2b^2m^2 = 0$$

$$-b^2m^2n^2 + b^4m^4 + a^2l^2b^2m^2 = 0$$

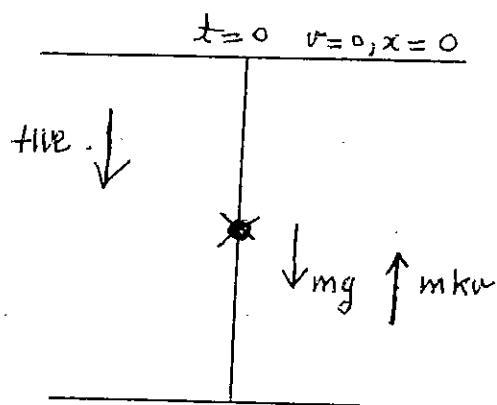
$$-m^2n^2 + b^2m^4 + a^2l^2m^2 = 0$$

$$-n^2 + b^2m^2 + a^2l^2 = 0$$

$$\text{ie } n^2 = a^2l^2 + b^2m^2$$

Question 7

(a)



$$F = ma$$

$$m\ddot{x} = mg - mkv$$

$$\ddot{x} = g - kv \quad \text{--- (1)}$$

(b) For terminal velocity w let $\ddot{x} = 0$

$$\therefore g - kw = 0$$

$$w = \frac{g}{k}$$

(c)

Now from (1)

$$\ddot{x} = k \left(\frac{g}{k} - v \right)$$

$$\ddot{x} = k(w - v) \quad \text{--- (2)}$$

(d) In (2) let $\ddot{x} = v \frac{dv}{dx}$

$$v \frac{dv}{dx} = k(w - v)$$

$$\frac{dv}{dx} = k \left(\frac{w - v}{v} \right)$$

$$\frac{dx}{dv} = \frac{1}{k} \times \left(\frac{v}{w - v} \right)$$

$$\frac{dx}{dv} = \frac{1}{k} \left[-1 - \frac{w}{v - w} \right]$$

Note

$$\frac{v}{w - v} = - \left(\frac{v}{v - w} \right)$$

$$= - \left(\frac{v - w + w}{v - w} \right)$$

$$= - \left(1 + \frac{w}{v - w} \right)$$

$$= -1 - \frac{w}{v - w}$$

(f) Now replacing $\ddot{x} = \frac{dv}{dt}$ in ②

$$(1) \quad \frac{dv}{dt} = k(w-v)$$

$$\frac{dt}{dv} = \frac{1}{k} \times \left(\frac{1}{w-v} \right) \quad \text{Note Several students used substitution here: let } u = w-v \text{ etc}$$

$$t = \frac{1}{k} \ln |w-v| + c$$

$$t = -\frac{1}{k} \ln |w-v| + c$$

Now when $t = 0$, $v = 0$

$$0 = -\frac{1}{k} \ln |w| + c$$

$$c = \frac{1}{k} \ln |w|$$

$$\therefore t = \frac{1}{k} \ln |w| - \frac{1}{k} \ln |w-v|$$

$$t = \frac{1}{k} \ln \left| \frac{w}{w-v} \right| \quad \rightarrow \quad \text{--- (4)}$$

Now 75% of terminal velocity means $v = 0.75w$.

$$t = \frac{1}{k} \ln \left| \frac{w}{w-0.75w} \right|$$

$$t = \frac{1}{k} \ln \frac{1}{0.25}$$

$$t = \frac{1}{k} \ln \frac{1}{\frac{1}{4}}$$

$$t = \frac{1}{k} \ln 4 \text{ seconds.}$$

ie the time to reach 75% of terminal velocity

Note
 $0.25 = \frac{1}{4}$

$$x = \frac{1}{k} \left[-v - w \ln|v-w| \right] + C = -\frac{v}{k} - \frac{w}{k} \ln|v-w| + C$$

Now $x=0$ when $v=0$

$$0 = -\frac{w}{k} \ln|-w| + C$$

$$C = \frac{w}{k} \ln|-w|$$

$$\therefore x = -\frac{v}{k} - \frac{w}{k} \ln|v-w| + \frac{w}{k} \ln|-w|$$

$$x = -\frac{v}{k} - \frac{w}{k} \left[\ln|v-w| - \ln|-w| \right]$$

$$x = -\frac{v}{k} - \frac{w}{k} \ln \left| \frac{v-w}{-w} \right|$$

$$x = -\frac{v}{k} - \frac{w}{k} \ln \left| \frac{w-v}{w} \right| \quad \text{--- (3)}$$

(e) Now when $x=H$, $v=w$ sub in

$$H = -\frac{u}{k} - \frac{w}{k} \ln \left| \frac{w-u}{w} \right|$$

$$kH = -u - w \ln \left| 1 - \frac{u}{w} \right|$$

$$w \ln \left| 1 - \frac{u}{w} \right| + u + kH = 0$$

$$\ln \left| 1 - \frac{u}{w} \right| + \frac{u}{w} + \frac{kH}{w} = 0$$

(ii) Back to equation (4)

$$t = \frac{1}{k} \ln \left| \frac{w}{w-v} \right|$$

$$kt = - \ln \left| \frac{w-v}{w} \right|$$

$$\ln \left| \frac{w-v}{w} \right| = -kt$$

$$\frac{w-v}{w} = e^{-kt}$$

$$w-v = we^{-kt}$$

$$v = w - we^{-kt}$$

$$\frac{dx}{dt} = w - we^{-kt}$$

$$x = wt + \frac{w}{k} e^{-kt} + c$$

Now when $x = 0$, $t = 0$

$$0 = 0 + \frac{w}{k} e^0 + c$$

$$c = -\frac{w}{k}$$

$$x = wt + \frac{w}{k} (e^{-kt} - 1)$$

When $t = \frac{1}{k} \ln 4$

$$x = \frac{w}{k} \ln 4 + \frac{w}{k} \left(e^{-k \times \frac{1}{k} \ln 4} - 1 \right)$$

$$x = \frac{w}{k} \left[\ln 4 + e^{-\ln 4} - 1 \right]$$

$$x = \frac{w}{k} \left(\ln 4 + e^{\ln 4^{-1}} - 1 \right)$$

$$x = \frac{w}{k} \left(\ln 4 + 4^{-1} - 1 \right)$$

$$x = \frac{w}{k} \left(\ln 4 - \frac{3}{4} \right) \text{ metres.}$$

(ii) Alternative approach:

Substitute $v = 0.75w$
in equation (3) part (i)

$$x = -\frac{0.75w}{k} - \frac{w}{k} \ln \left(\frac{w-0.75w}{w} \right)$$

$$x = -\frac{w}{k} \left[0.75 + \ln \left(\frac{w(1-0.75)}{w} \right) \right]$$

$$x = -\frac{w}{k} \left[\frac{3}{4} + \ln 0.25 \right]$$

$$x = -\frac{w}{k} \left(\frac{3}{4} + \ln \frac{1}{4} \right)$$

$$x = -\frac{w}{k} \left(\frac{3}{4} + \ln 4^{-1} \right)$$

$$x = -\frac{w}{k} \left(\frac{3}{4} - \ln 4 \right)$$

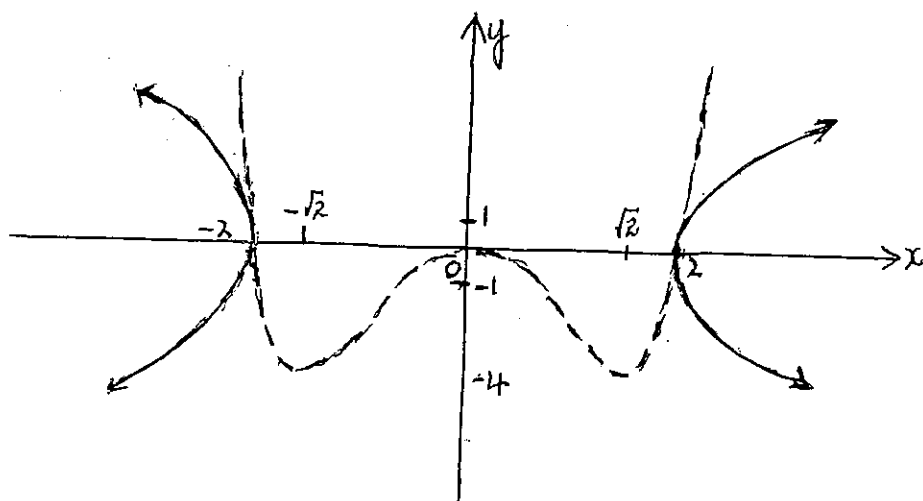
$$x = \frac{w}{k} \left(\ln 4 - \frac{3}{4} \right) \text{ metres.}$$

This is the distance fallen at 75% of the terminal velocity

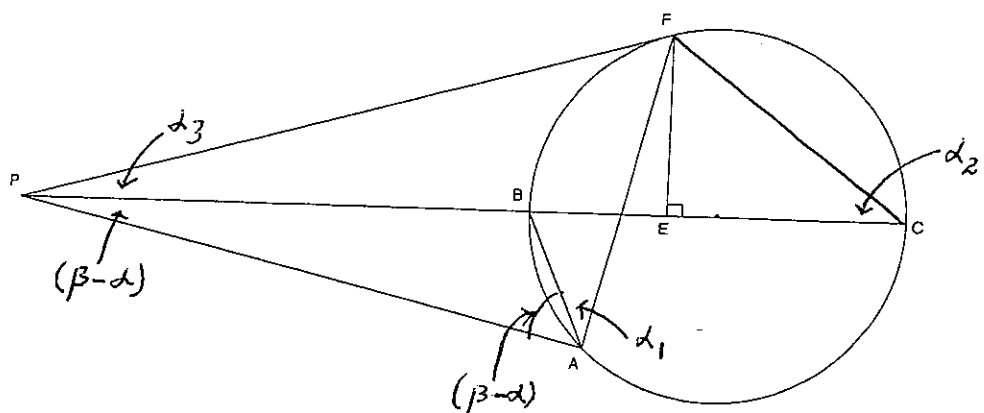
Question 8

(a) $y^2 = x^2(x^2 - 4)$

Dotted Graph: $g(x) = x^2(x^2 - 4)$



(b)



(i) arc $AF = \text{arc } FC$ {given} $\therefore AF = FC$ --- ①

But in $\triangle PFC$, EF perpendicularly bisects base PC .

$\therefore \triangle PFC$ is isosceles {converse of The Theorem for an isosceles $\triangle$ which states that the altitude is the perpendicular bisector of the base}

$\therefore PF = FC$ --- ②

From ① and ② $AF = PF$ $\therefore \triangle APF$ is isosceles.

(ii) Now $d_1 = d_2$ {Angles in the same segment Theorem - BF subtends equal angles at A and C }

And $d_2 = d_3$ {base angles of isosceles $\triangle PFC$ }

Now since $\triangle APF$ is isosceles

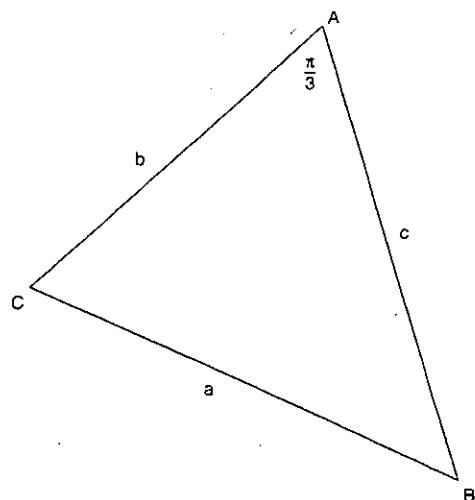
$\angle FPA = \angle FAP = \beta$, say

$\therefore \angle BPA = \angle BAP = \beta - \alpha$

$\therefore \triangle PBA$ is isosceles and $\therefore AB = BP$ --- ③

Hence, $EC = EP = EB + BP = EB + AB$ (from ③) $= AB + BE$.

(c)



(i)

using the cosine rule :

$$a^2 = b^2 + c^2 - 2bc \cos \frac{\pi}{3}$$

$$\therefore a^2 = b^2 + c^2 - 2bc \left(\frac{1}{2}\right)$$

$$\therefore a^2 = b^2 + c^2 - bc$$

Now $a^2 - bc = b^2 + c^2 - 2bc$ (subtracting bc from both sides)

$$\therefore a^2 - bc = (b - c)^2$$

Now $(b - c)^2$ is a perfect square

$$\therefore a^2 - bc \geq 0 \quad (\text{note equality if } b = c)$$

$$\therefore a^2 \geq bc \quad \text{--- ①}$$

(iii) The area of $\triangle ABC = \frac{1}{2} bc \sin \frac{\pi}{3}$

$$\text{Area} = \frac{1}{2} bc \times \frac{\sqrt{3}}{2}$$

$$\text{Area} = \frac{\sqrt{3}}{4} bc$$

$$\text{Area} \leq \frac{\sqrt{3}}{4} a^2 \quad (\text{using part (i) ①})$$